

Boots in the Beat

Effects of Beat Policing on Victimization and Crime Perception

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Outline

- 1 Motivation
- 2 Institutional Setting
- 3 Data and Empirical Strategy
- 4 Main Results
- 5 Spillovers
- 6 Heterogeneity
- 7 What Is Driving the Effects?
- 8 Conclusion

- Crime is a top public concern and a major driver of public and private spending worldwide.
- We know a lot about **how many** police resources reduce crime.
- We know much less about **how those resources are deployed**.
 - Same number of officers, organized differently across space, may perform very differently.
- **Beat policing**: permanently assigning officers to small, fixed territorial units, rather than rotating them across a larger area.
 - Widely adopted around the world.
 - Rigorous causal evidence on its effects remains scarce.

- We study *Plan Cuadrante*, a large-scale beat-policing reform in Chile.
- It subdivided municipalities into smaller **quadrants** and permanently assigned officers to them.
- We exploit the **staggered rollout** of the program (1998–2013) across Chilean municipalities.
- Rich administrative **and** household survey data let us look at:
 - Actual crime (victimization, reported crime, arrests)
 - Perceived security (crime perception, trust, private security investment)
- We separate the **deployment/strategy** channel from the **resources** channel.

Crime and safety

- Victimization: -10 pp (36%)
- Reported crime: -10% (208 per 100k)
- Arrests: $+14\%$ (48 per 100k)
- Effects hold in municipalities with **no differential resource increase**.
- No evidence of spatial spillovers to neighboring municipalities.

Perceptions and behavior

- Crime perception: -15 pp (36%)
- Private security investment: -8 pp (37%)
- Trust in the police: $\uparrow$

Contribution to the Literature

- ① First causal evidence on **beat policing** at scale, on both crime *and* security perceptions.
- ② Complements the police-organization literature (response times, prioritization, technology) by studying **territorial deployment** as a distinct margin.
- ③ Complements the police-resources literature: **how** resources are organized matters, not only **how much** there is.

- National, hierarchical police force; centrally defined doctrine, limited local discretion.
- Basic operational unit: the *comisaría* — historically covered the **entire municipal territory**.
- Average *comisaría*: 1,608 km², 57,642 inhabitants.

Plan Cuadrante(Plan Cuadrante de Seguridad Pública)

- Rolled out 1998–2013 in municipalities with $\geq 70\%$ urban population (150 eligible municipalities).
- Subdivided each *comisaría* into ~ 7 **quadrants** on average.
- Officers **permanently** assigned to one quadrant (vs. rotating across the whole municipality).
- Average quadrant: 185 km², 13,466 inhabitants, ~ 3 officers per shift.
- National operational guidelines unchanged $\Rightarrow$ strategy adjustments emerge *endogenously* from the new deployment, not from a top-down redesign.

Rollout of *Plan Cuadrante*

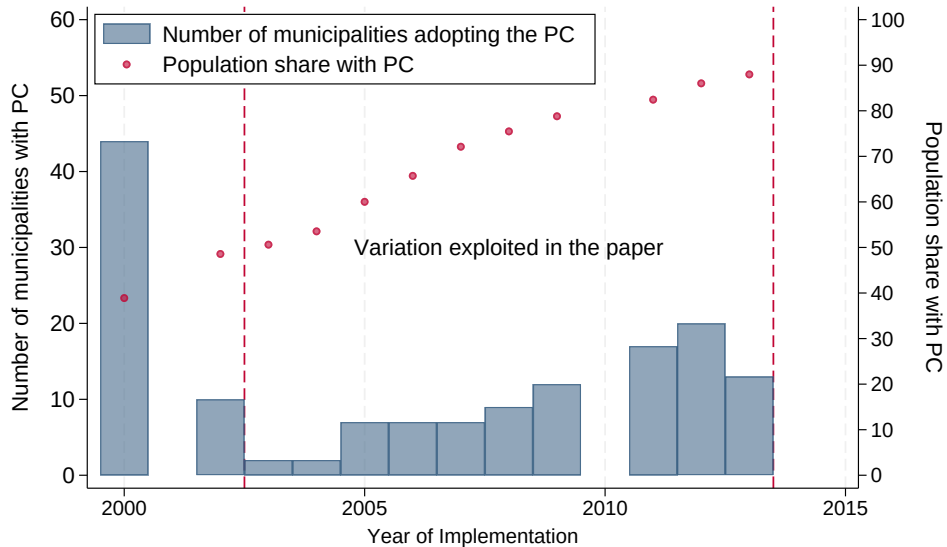
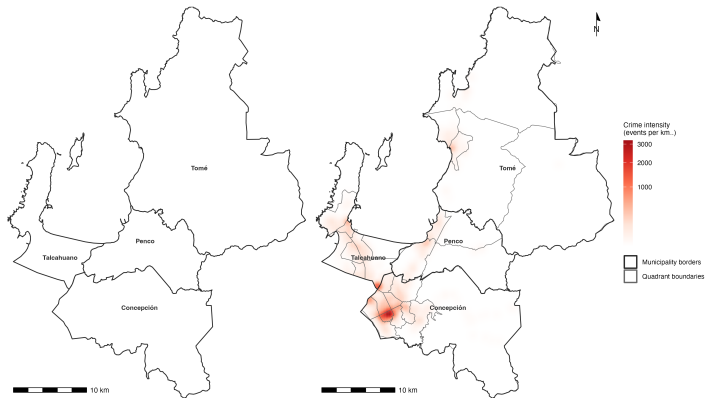


Illustration: Quadrants in the Concepción Metropolitan Area



ENUSC survey (INE)

- Repeated cross-section, 25,000+ households/year, 101 urban municipalities.
- Victimization, perceptions, trust, private security investment.
- Sample: 46 treated (2005–2012) + 1 never-treated municipality.
- ~27% of Chile's population.

Administrative records

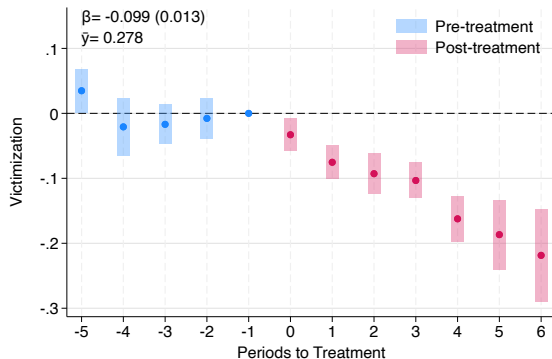
- Reported crime (2002–2016) and arrests (2005–2016), *Undersecretary for Crime Prevention*.
- All Chilean municipalities.
- Sample: 96 treated (2003–2013) + 196 never-treated.
- Over half of Chile's population.

$$Y_{imt} = \beta_0 + \beta_1 PC_{mt} + \mu_m + \mu_t + \varepsilon_{imt}$$

- Staggered adoption $\Rightarrow$ canonical two-way fixed effects can generate **negative weights** (Goodman-Bacon).
- We use **Callaway and Sant'Anna (2021)**'s estimator for staggered DiD.
- Robust to alternative staggered-DiD estimators (de Chaisemartin–D'Haultfoeuille, Gardner, Wooldridge); canonical TWFE attenuates, as expected.
- Standard errors clustered at the municipality level, wild bootstrap (few, large clusters).

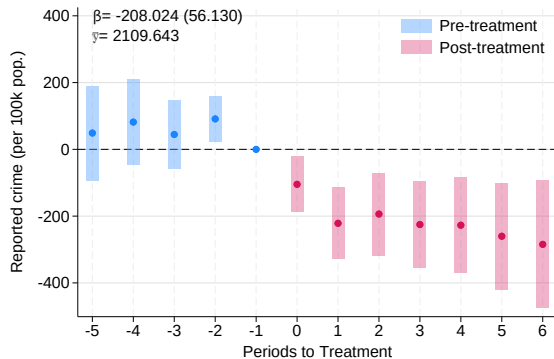
- **Key assumption:** parallel trends absent *Plan Cuadrante*.
- Pre-treatment coefficients are small and jointly indistinguishable from zero for all four main outcomes.
 - p -values (avg. pre-trend = 0): victimization 0.833; reported crime 0.204; crime perception 0.178; security measures 0.066.
- **Two main threats:**
 - ① Spatial spillovers (Section 5) → tested directly, no evidence found.
 - ② Resource increases confounded with deployment (Section 7) → addressed via heterogeneity analysis.

Victimization and Reported Crime



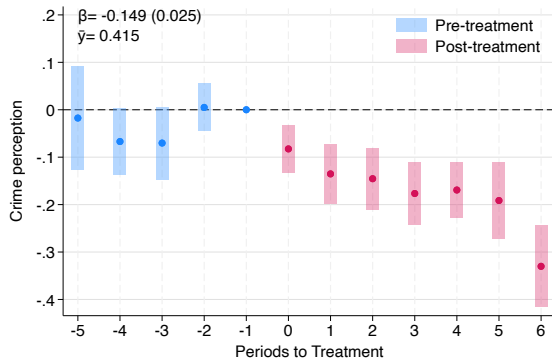
(a) Victimization (ENUSC)

● Victimization: -10 pp (36%). Reported crime: -208 per 100k (10%).

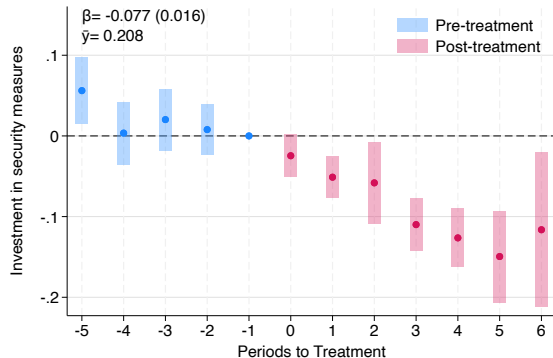


(b) Reported crime (admin.)

Crime Perception and Private Security Investment



(c) Perceived increase in crime



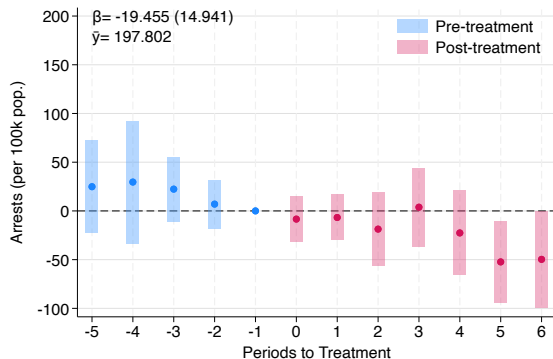
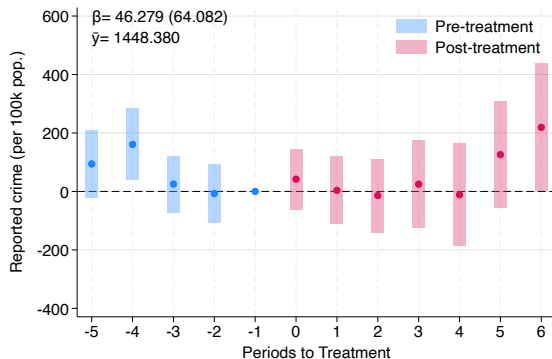
(d) Household security measures

- Crime perception: -15 pp (36%). Security investment: -8 pp (37%).
- **Crime and perceived risk moved together** — not always the case in the literature.

Are These Effects Really Spillover-Free?

- Displacement to neighboring municipalities would bias our estimates **upward**.
- *A priori*, spillovers are less likely here:
 - *Plan Cuadrante* reorganizes the *entire* municipal territory, not just crime hot spots.
 - Municipalities are large (avg. 101,000 inhabitants, 2,000 km²).
 - Officers are tied to *comisarías* for fixed periods $\Rightarrow$ limited rapid reallocation.
- We test this directly with **three** complementary exercises.

No Evidence of Spillovers to Untreated Neighbors

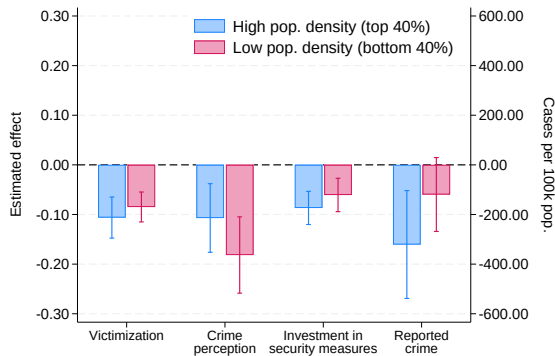


Reported crime: neighbors of adopters

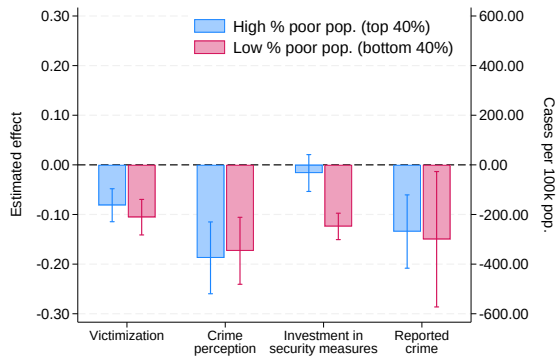
Arrests: neighbors of adopters

- Also: continuous exposure (share/number of treated neighbors) and a placebo/permutation exercise — all null.

Effects Are Broadly Similar Across Municipalities



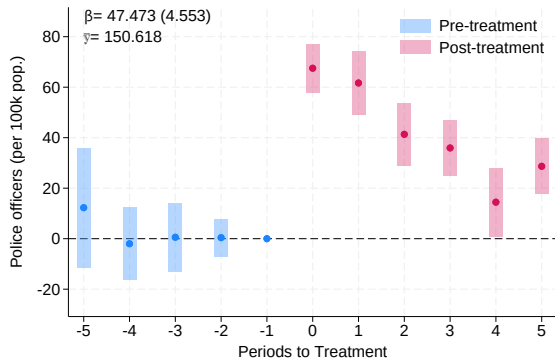
(a) By population density



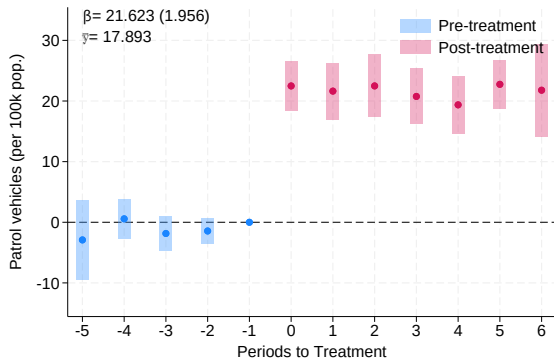
(b) By poverty rate

- Also similar by household socioeconomic status.
- **External validity:** gains are not confined to a particular type of municipality or household.

Confound: *Plan Cuadrante* Also Came with More Resources



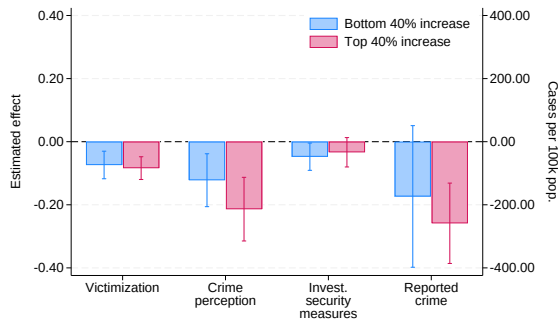
(a) Police officers



(b) Patrol vehicles

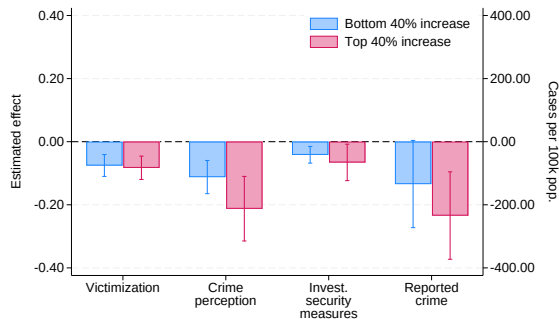
- Top-40% resource-increase municipalities: officers +22%, vehicles +80%, relative to control-like municipalities.
- **Question:** are our effects just a resources story?

Effects Hold Even Without Differential Resource Increases



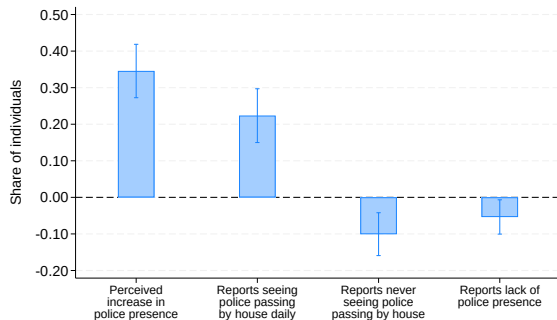
(e) By intensity of Δ officers

- Bottom-40% resource-increase municipalities (blue) — resource growth similar to control group — still show sizeable effects.
- Complementary evidence: *Seguridad Ciudadana* (resources w/o deployment change) shows no effect; resource variation *within* already-treated municipalities is unrelated to outcomes.

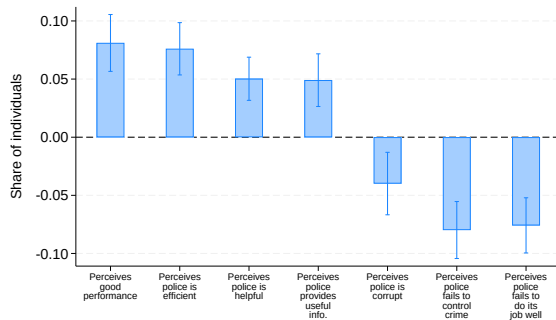


(f) By intensity of Δ patrol vehicles

Police Presence and Perceived Performance



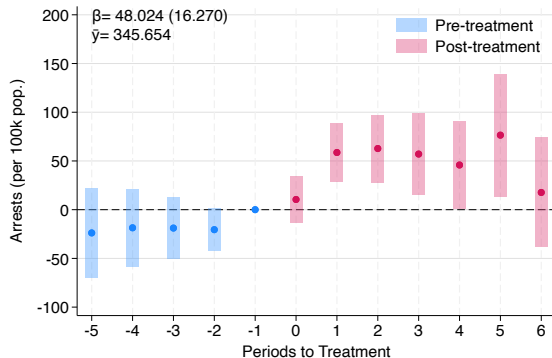
(a) Perceived police presence



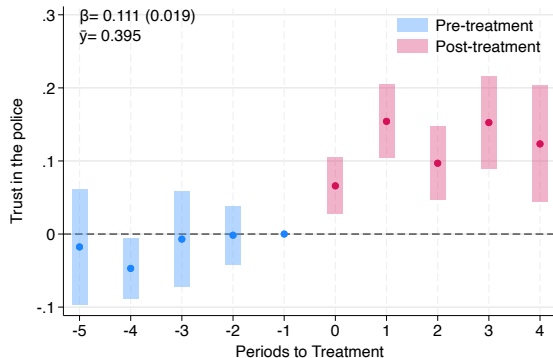
(b) Perceived police performance

- Perceived police presence: +35 pp (156%). Police rated as good/efficient/helpful/informed $\uparrow$; corrupt/ineffective $\downarrow$.

Arrests and Trust: Effectiveness, Not Just Presence



(c) Arrests (per 100k)



(e) Trust in the police

- Arrests +14%; arrest/reported-crime ratio +1.7 pp (10%) — **while crime is falling.**
- Consistent with officers building **local knowledge** under stable territorial assignment, not merely patrolling more.

- Resources were likely important for **implementation**, and may explain part of the effect in some places.
- But resource increases **alone** cannot explain our findings:
 - Effects hold where resource growth resembled the control group.
 - A pure-resources reform (*Seguridad Ciudadana*) shows no effect.
 - Arrests rise *while crime falls* — hard to reconcile with more patrolling alone.
- The evidence points to **permanent territorial assignment and local knowledge** as key drivers.

- **Beat policing** — permanent, small-scale territorial assignment — generates large gains in public safety **and** security perceptions.
- Crime and perceived risk fell **together**, avoiding the over-investment in private security that misaligned risk perceptions can cause.
- Trust in the police rose — notoriously hard to achieve in middle-income settings.
- The gains are **not** simply bought with more resources: **how** police are deployed matters.

When Should We Expect Beat Policing to Work?

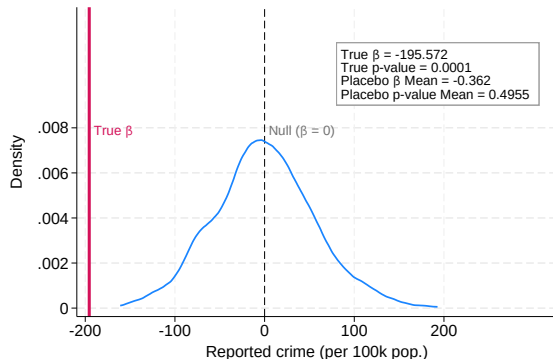
- *Plan Cuadrante* operated in a specific institutional context:
 - ① Low organized-crime presence $\Rightarrow$ officers could stay in an area without intimidation risk.
 - ② A professionally trained police force $\Rightarrow$ capacity to implement the reform well.
- Neither feature is unique to Chile — both are present, or attainable, in many settings.
- **Policy takeaway:** reorganizing *deployment* can be a cost-effective complement (or alternative) to simply adding more police resources.

Thank you!

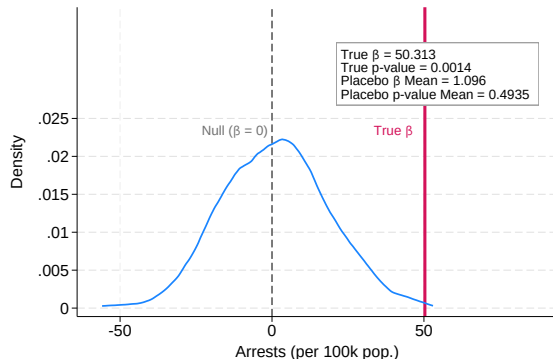
Comments and questions welcome.

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Backup: Randomization-Inference Placebo Test



Reported crime



Arrests

- Placebo distribution from 1,000 random reassignments of adoption dates/never-treated status.
- True direct effects sit far in the tail of the placebo distribution.

Backup: Additional Robustness Checks

- Alternative staggered-DiD estimators (de Chaisemartin–D'Haultfœuille; Gardner; Wooldridge): similar magnitude and significance.
- Balanced panel of municipalities: nearly identical estimates.
- Full (uncapped) event-study window: nearly identical estimates.
- Bonferroni correction for multiple hypothesis testing: all main results remain significant.
- Excluding municipalities affected by the 2010 Maule earthquake: nearly identical estimates.
- Placebo test on non-neighborhood crime (e.g., sexual abuse): no effect, as expected.

Backup: Key References

- Callaway, B. and Sant'Anna, P. (2021). "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics*.
- Goodman-Bacon, A. (2021). "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics*.
- Blattman, C., Green, D., Ortega, D., and Tobón, S. (2021). "Place-Based Interventions at Scale." *Journal of the European Economic Association*.
- Mastrobuoni, G. (2020). "Police Disruption and Performance." *Journal of Public Economics*.